

Singularity Analysis of ABB's GoFa 5 Robot Arm

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Abstract

In the past decade, manipulator arms with non-traditional architectures—once found mainly in space and painting applications—have become popular as collaborative robots. Examples include ABB's YuMi and GoFa, Kinova's Link 6, and Fanuc's CRX. These cobots lack closed-form inverse kinematics solutions, making it impossible to unambiguously select one configuration among the 16 (or infinitely many) that correspond to a given end-effector pose, which may create safety risks. Moreover, they exhibit far more singularities than typical manipulators, and most of them are far more complex to describe. Nevertheless, many authors argue these manipulators can provide improved dexterity and a larger workspace. In this paper, we analyze the singularities of ABB's GoFa using Grassmann line geometry and provide straightforward, sufficient (though conservative) conditions for avoiding them. Then, while GoFa can exhibit over a dozen distinct singularities compared to only three (wrist, shoulder, and elbow) for traditional robot arms, we attempt to quantify which architecture actually possesses a greater amount of singular and near-singular configurations.

Methodology

Table 1: DH table of ABB's GoFa 5

i	θ_i	d_i (mm)	a_i (mm)	α_i
1	θ_1	265	0	-90°
2	$\theta_2 - 90^\circ$	0	444	0°
3	θ_3	0	110	-90°
4	θ_4	470	0	90°
5	θ_5	0	80	-90°
6	$\theta_6 + 180^\circ$	101	0	0°

The Jacobian matrix is defined as:

$$\mathbf{J} = [\$1 \quad \$2 \quad \$3 \quad \$4 \quad \$5 \quad \$6],$$

where each column is a screw

$$\$i = \begin{bmatrix} \mathbf{e}_i \\ \mathbf{e}_i \times \mathbf{r}_i \end{bmatrix}.$$

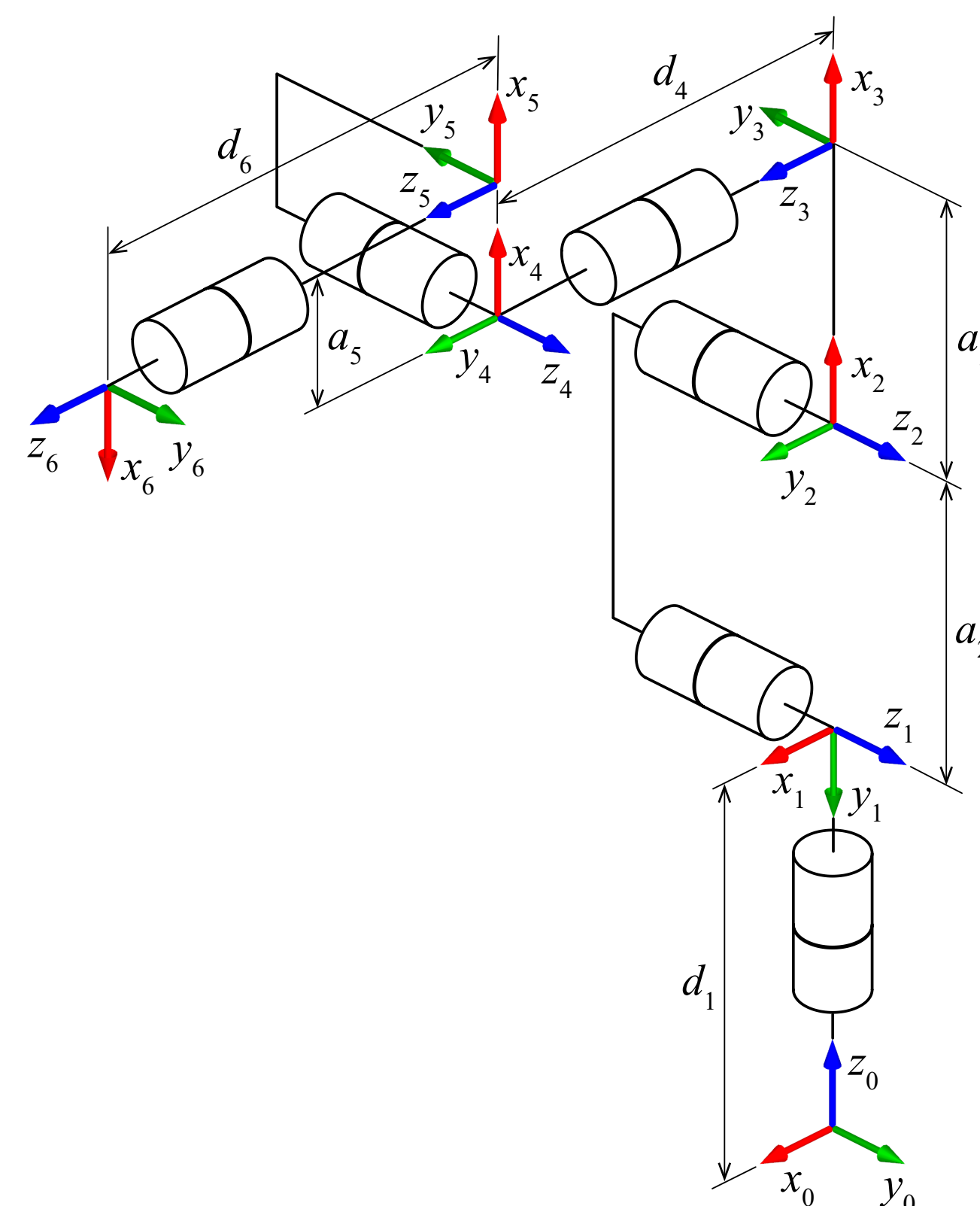


Figure 1: Schematic of ABB's GoFa 5

$$\mathbf{J} = \begin{bmatrix} c_{23} & 0 & 0 & 0 & s_4 & -c_4 s_5 \\ 0 & -1 & -1 & 0 & -c_4 & -s_4 s_5 \\ -s_{23} & 0 & 0 & 1 & 0 & c_5 \\ 0 & a_2 s_3 & 0 & 0 & d_4 c_4 & s_4(d_4 s_5 + a_5) \\ -a_3 s_{23} - a_2 s_2 & 0 & 0 & 0 & d_4 s_4 & -c_4(d_4 s_5 + a_5) \\ 0 & a_2 c_3 + a_3 & a_3 & 0 & 0 & 0 \end{bmatrix}$$

Results

We formulate the problem of finding dependent screws as a MATLAB optimization problem and relate each singular configuration to a geometric interpretation following [1].

Table 2: Singularities of GoFa 5

Case	Axes	Condition	Loci in $\mathcal{Q}$, 4D space of joints 2–5	Examples of joint angles
A_1	$\mathcal{L}_1, \mathcal{L}_4$	$\mathcal{L}_1 \equiv \mathcal{L}_4$	2 planes, total area of 79 rad ²	$0^\circ, 14.34^\circ, -104.34^\circ, -10^\circ, 80^\circ, 0^\circ$
A_2	$\mathcal{L}_1, \mathcal{L}_6$	$\mathcal{L}_1 \equiv \mathcal{L}_6$	4 curves, total length of 51 rad	$0^\circ, -30^\circ, -8.1^\circ, 0^\circ, -51.9^\circ, 0^\circ$
B_1	$\mathcal{L}_1, \mathcal{L}_4, \mathcal{L}_5$		2 surfaces, total area of 157 rad ²	$0^\circ, -30^\circ, -19.45^\circ, -90^\circ, -80^\circ, 0^\circ$
B_2	$\mathcal{L}_1, \mathcal{L}_4, \mathcal{L}_6$		4 surfaces, total area of 297 rad ²	$0^\circ, 10^\circ, 10^\circ, 0^\circ, -7.77^\circ, 0^\circ$
B'_2	$\mathcal{L}_1, \mathcal{L}_4, \mathcal{L}_6$		8 lines, total length of 61 rad	$0^\circ, 40^\circ, 50^\circ, 0^\circ, 0^\circ, 0^\circ$
B_3	$\mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_5$		2 planes, total area of 79 rad ²	$0^\circ, 20^\circ, -76.83^\circ, 0^\circ, 60^\circ, 0^\circ$
B_4	$\mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_6$		4 lines, total length of 25 rad	$0^\circ, 20^\circ, -78.69^\circ, 90^\circ, 90^\circ, 0^\circ$
C_1	$\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_4, \mathcal{L}_6$		2 planes, total area of 79 rad ²	$0^\circ, 20^\circ, -104.34^\circ, 15^\circ, -174.9^\circ, 0^\circ$
D_1	$\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5$		2 planes, total area of 79 rad ²	$0^\circ, 0^\circ, -76.83^\circ, 15^\circ, 25^\circ, 0^\circ$
D_2	$\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_6$		2 surfaces, total area of 199 rad ²	$0^\circ, 0^\circ, 40^\circ, 15^\circ, -8.18^\circ, 0^\circ$
D'_2	$\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_6$		16 points	$0^\circ, 90^\circ, 0^\circ, -90^\circ, -9.8^\circ, 0^\circ$
E_1	$\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5, \mathcal{L}_6$		4 surfaces, total area of 397 rad ²	$0^\circ, -30^\circ, 30^\circ, -90^\circ, -8.63^\circ, 0^\circ$

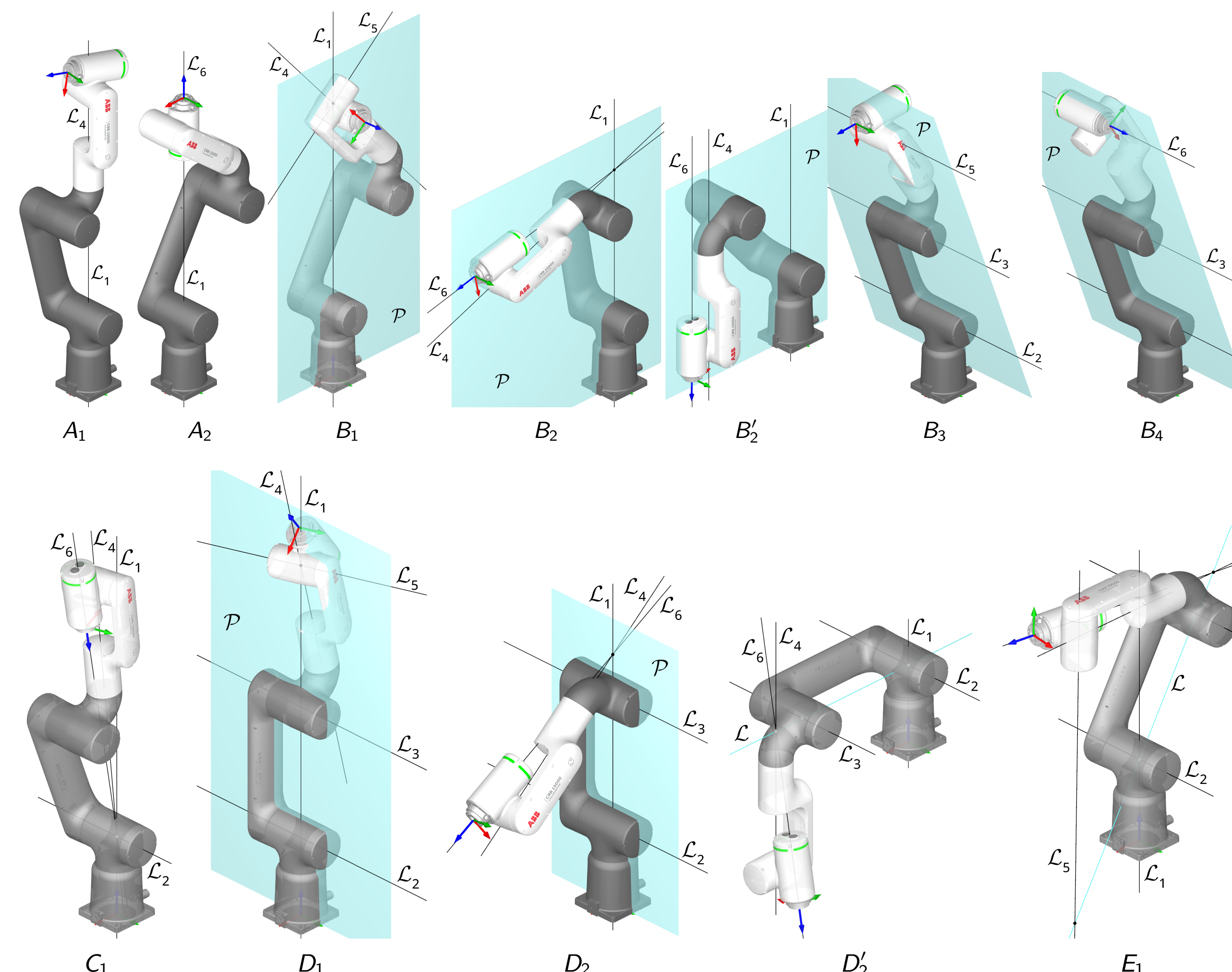


Figure 2: Geometric interpretation of kinematic singularity present in GoFa 5

Table 3: Sufficient conditions for avoiding singularities in GoFa 5

Simplified conditions to avoid	Possible singularities
$\theta_2 = 0^\circ$ or 180°	D_1 and D_2
$\theta_3 = -104.34^\circ$	A_1 and C_1
$\theta_4 = 0^\circ$ or $\pm 90^\circ$ or 180°	$A_2, B_1, B_2, B'_2, B_3, B_4, D'_2$, and E_1

Comparison with A0509 and UR5e

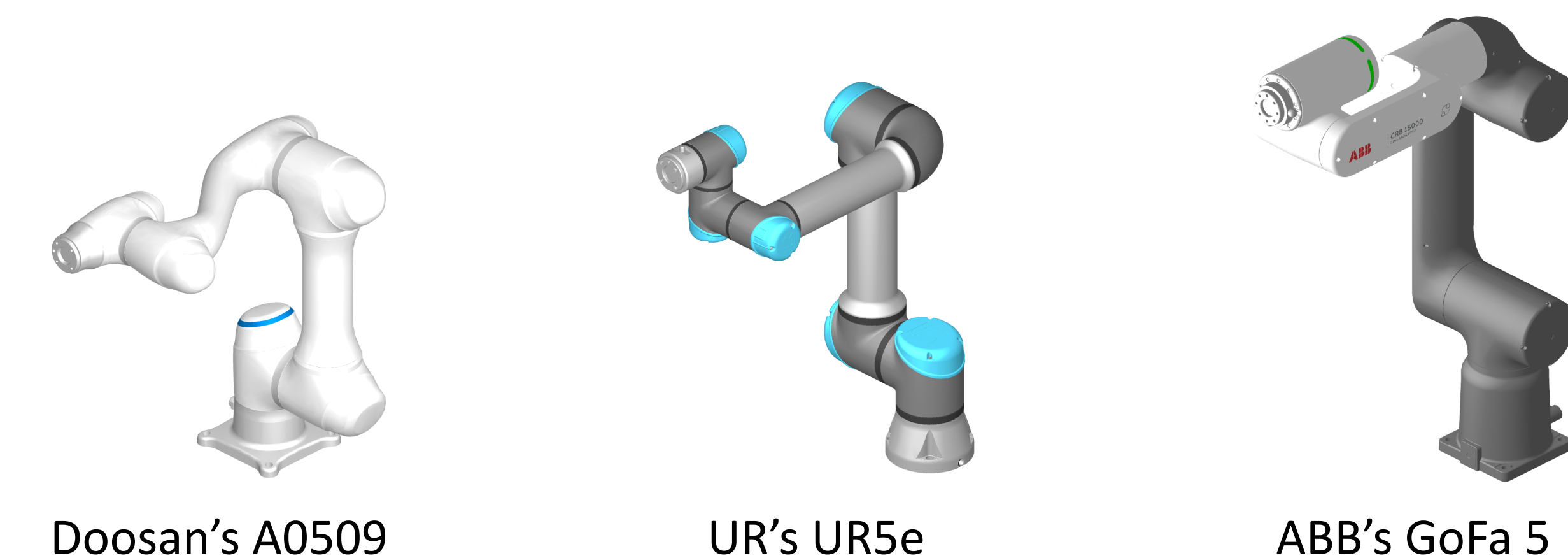


Figure 3: Three distinct robot arm architectures

Singularities of the A0509 and UR5e are described in [2] and [4]; however, their morphologies have not yet been compared.

Table 4: Singularities of A0509

Case	Axes	Condition
A_1	$\mathcal{L}_4, \mathcal{L}_6$	$\mathcal{L}_4 \equiv \mathcal{L}_6$
C_1	$\mathcal{L}_1, \mathcal{L}_4, \mathcal{L}_5, \mathcal{L}_6$	
E_1	$\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5, \mathcal{L}_6$	

Table 5: Singularities of UR5e

Case	Axes	Condition
B_1	$\mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4$	
C_1	$\mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_6$	
E_1	$\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5, \mathcal{L}_6$	

Manipulability extends singularities by quantifying how freely the end-effector can move and is defined as the volume of the manipulability ellipsoid [5]:

$$\mu = \sqrt{\det(\mathbf{J}\mathbf{J}^T)}.$$

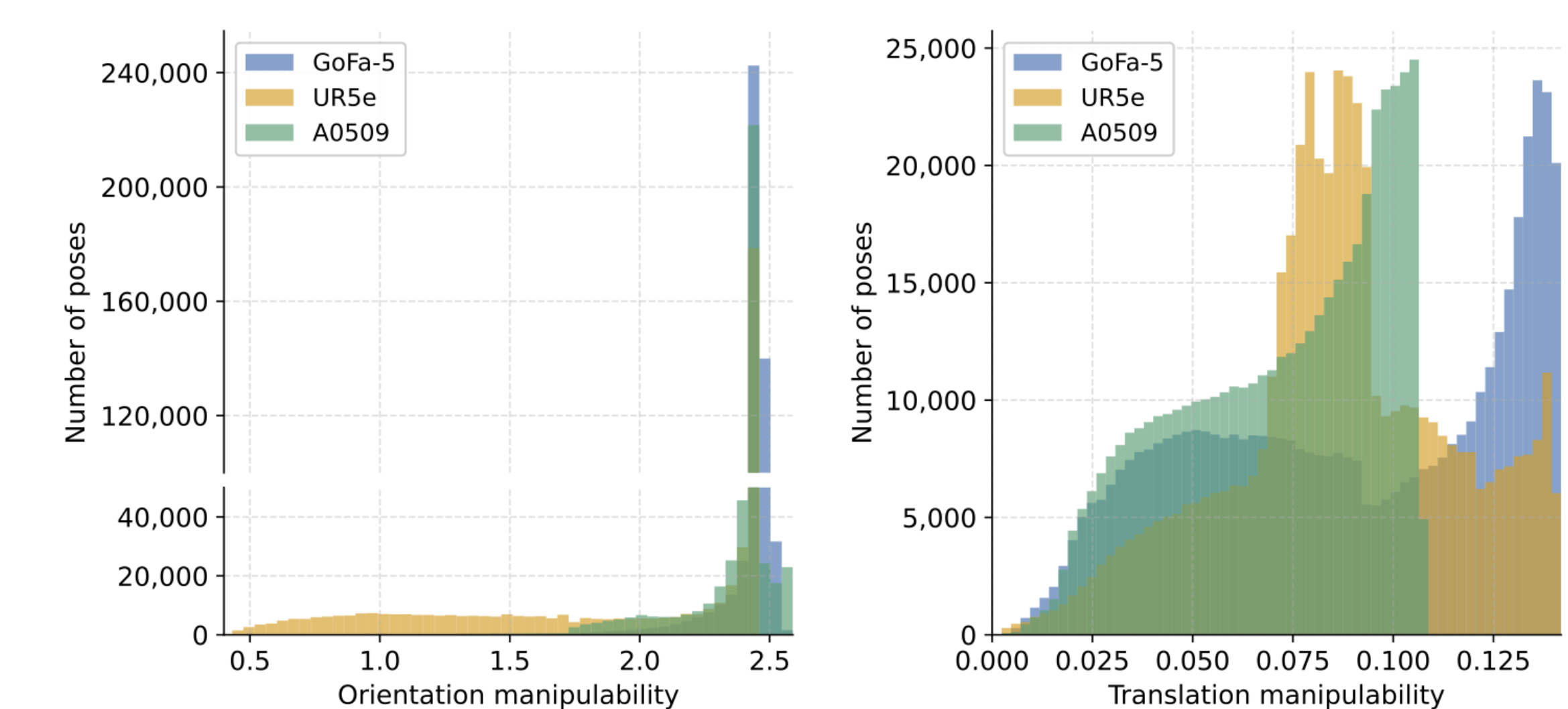


Figure 4: Manipulability across specified Cartesian workspace

References

- [1] Hao et al., *Conditions for line-based singularities in spatial platform manipulators*, 1998.
- [2] Hayes et al., *Singular configurations of wrist-partitioned 6R serial robots*, 2002.
- [3] Weyrer et al., *Singularity avoidance of a mobile manipulator*, 2019.
- [4] FarzanehKaloorazi & Bonev, *Singularities of the typical collaborative robot arm*, 2018.
- [5] Yoshikawa, *Dynamic manipulability of robot manipulators*, 1985.